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Spatial Variation of Supply Water Quality Parameters Across Dhaka City

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Abstract

In urban areas like Dhaka, point-of-use water quality monitoring is often overlooked due to the high cost and difficulty of widespread sampling. This study evaluates the spatial variation of key water quality parameters hardness, sulfate, chloride, and nitrate ions in Dhaka Water Supply and Sewerage Authority (DWASA) supplied tap water across 13 locations in Dhaka city. Water samples were collected directly from household taps to represent actual consumer exposure. The analysis revealed notable variations among locations. Amin Bazar recorded the highest water hardness (300 mg/l), while Kuril showed the lowest (33 mg/l). Chloride levels were highest in Amin Bazar (233.93 mg/l) and lowest in Rampura (39.99 mg/l). Sulfate ion concentrations were mostly below 20 mg/l, except in Amin Bazar (90 mg/l) and Kamlapur (42 mg/l), indicating local anomalies. Nitrate was detected only in Tongi (15 mg/l) and Amin Bazar (20 mg/l), with all other locations showing 0 mg/l. These variations suggest the influence of pipe age, source water quality, and potential infrastructure issues. The study emphasizes the need for regular and localized monitoring to help identify areas with deteriorating pipelines and support Water Supply Authorities (WSA) in prioritizing maintenance efforts effectively.

Keywords: *Dhaka Water Supply and Sewerage Authority (DWASA), Water Supply Authorities (WSA)*

1 Introduction

Access to clean and safe drinking water is essential for human health, well-being, and sustainable urban development (E. Cicia et al., 2010; Crase et al., 2005). In densely populated cities like Dhaka, the capital of Bangladesh, the provision of safe and reliable tap water has become increasingly challenging due to rapid urbanization, industrial expansion, population growth, and aging infrastructure (Cidu et al., 2010). With over 10.28 million residents Dhaka faces mounting pressure on its municipal water supply system, resulting in widespread concerns over water safety and quality. (Bangladesh Bureau of Statistics [BBS], 2022). Water contamination in urban settings can stem from several factors, including microbial pathogens, heavy metals, and chemical pollutants, often aggravated by poor water treatment practices, outdated pipelines, and illegal industrial discharges (Allaire et al., 2018; Trueman et al., 2019). Numerous studies have shown that insufficient oversight and lack of maintenance in water distribution systems can lead to significant deterioration in tap water quality, heightening the risk of gastrointestinal illnesses, heavy metal toxicity, and other waterborne diseases (Cidu et al., 2010; Kachel et al., 2007; Zivin et al., 2011). Furthermore, public perception plays a critical role in water use behavior. Consumers often assess water quality based not only on objective safety parameters but also on sensory attributes like taste, odor, and color (Um et al., 2002; McDonald and Rundle, 2006). In Dhaka, this dis-connect between perceived and actual water quality has led to a growing dependence on bottled water and home filtration systems, reflecting broader issues of trust in public water utilities and regulatory enforcement (Crase, 2007; Zivin et al., 2011). Globally, the United Nations Sustainable Development Goals (SDGs) emphasize the importance of achieving universal and equitable access to safe and affordable drinking water by 2030 (UN, 2015). However, in many low- and middle-income urban centers like Dhaka, progress remains constrained by infrastructural, financial, and governance limitations. This study aims to evaluate both the objective and perceived quality of tap water in selected areas of Dhaka city. It compares municipal and bottled water sources in light of national and international drinking water standards, while also examining public perception through household-level surveys. The findings are intended to inform policy strategies for improving water quality governance, enhancing infrastructure, and protecting public health in rapidly urbanizing environment.

2.Methodology

2.1 Location Selection and collection of water Sample

According to the Bangladesh Bureau of Statistics' 2022 Census and official metropolitan records, Dhaka North City Corporation covers approximately 196.22 km², and Dhaka South City Corporation covers approximately

109.24 km², resulting in a combined administrative area of about 305.46 km² (Bangladesh Bureau of Statistics [BBS], 2022). This makes Dhaka the largest city in Bangladesh by area. The Dhaka Water Supply and Sewerage Authority (DWASA) is responsible for supplying water throughout the entire city. To assess selected water quality parameters, this study identified 13 locations across Dhaka City, encompassing both the North and South city corporations. These locations are illustrated in Figure 1. For the purpose of this study, household tap water samples were collected from each of the selected locations, as shown in Figure 2.

2.2 Determination of Water Quality Parameters

2.2.1 Chloride Ion (Cl^-)

2.2.1.1 Principle of the Test (Mohr's Method)

After collecting the water samples, they were transported to the laboratory and analyzed within 24 hours to ensure accuracy in water quality assessment. For the determination of chloride ions, a titration method using a standard silver nitrate ($AgNO_3$) solution was employed. The 0.0141 N $AgNO_3$ solution was prepared by accurately dissolving 2.395 g of silver nitrate in distilled water, transferring it to a 1-liter volumetric flask, and diluting it to the mark, as depicted in Fig. 3a. Since $AgNO_3$ is photosensitive, it was stored in a dark bottle. A 5% potassium chromate (K_2CrO_4) solution was used as an indicator, prepared by dissolving 5g of K_2CrO_4 in 100 mL of distilled water, then filtered and stored in a sealed container as depicted in Fig. 3b. A standard sodium chloride (NaCl) solution of 0.0141 N was also prepared by dissolving 0.824 g of pre-dried NaCl (at 105°C) in distilled water and diluting it to 1 liter as depicted in Fig. 3c. For the titration, 25 mL of the water sample was pipetted into a clean conical flask, followed by the addition of 1 mL of potassium chromate indicator. The sample was titrated with the silver nitrate solution until the appearance of a permanent red-brown color indicated the endpoint, as depicted in Fig. 3d, resulting from the formation of silver chromate (Ag_2CrO_4) after all chloride ions had reacted with silver to form silver chloride (AgCl). The volume of $AgNO_3$ consumed was recorded for chloride concentration calculation. The concentration of chloride ions (in mg/L) was calculated using the following formula:

$$\text{Chloride (mg/L)} = (V \times N \times 35.45 \times 1000)/\text{Sample Volume} \quad (1)$$

Where, V = volume of $AgNO_3$ used (mL), N = normality of $AgNO_3$, and 35.45 = molar mass of Cl^-

2.2.2 Hardness Determination

For total hardness determination, a clean 5 mL plastic beaker was rinsed and filled with the water sample. Five drops of hardness buffer and one drop of Calmagite indicator were added through the cap port, and the solution was gently swirled. A red-violet color indicated the presence of hardness ions. Using a titration syringe, EDTA titrant (HI 3812-0) was added dropwise with swirling after each drop. The color shifted from red-violet to purple, and finally to blue as shown in Fig. 4, marking the endpoint. The volume of EDTA used was recorded and multiplied by 300 to express total hardness in mg/L as $CaCO_3$.

2.2.3 Sulfate Ion Determination

Sulfate concentration in the water sample was determined using a visual turbidity-based method. A plastic vessel was filled with 50 mL of the sample, followed by the addition of one packet of HI 38000A-0 Sulfate Reagent, and the solution was gently swirled to ensure complete dissolution. Two drops of Complexing Agent and two spoons of HI 38000B-0 Reagent were then added with thorough mixing. The mixture was allowed to stand undisturbed for 5 minutes to allow for full reaction development. The test tube was placed on a white surface, and the sample was added dropwise using a plastic pipette until the black spot at the bottom was no longer visible as depicted in Fig. 5A and Fig. 5B. The sulfate concentration was determined by reading the liquid level in the tube, expressed in mg/L (ppm).

2.2.4 Nitrate Ion Determination

Nitrate concentration in the water sample was determined using a colorimetric method. A plastic vessel was rinsed and filled with 10 mL of the sample, after which one packet of HI 3834-0 reagent was added. The vessel was capped and shaken until the reagent fully dissolved. The resulting solution was transferred into a color comparator cube and left undisturbed for 4 minutes to allow complete color development. The final color was then visually compared with a standard reference scale as shown in Fig. 6, and the corresponding concentration was recorded in mg/L (ppm). A noticeable color change in the sample collected from Amin Bazar confirmed the presence of nitrate in the water. This colorimetric response confirms that nitrate contamination is present at measurable levels in the Amin Bazar water sample.

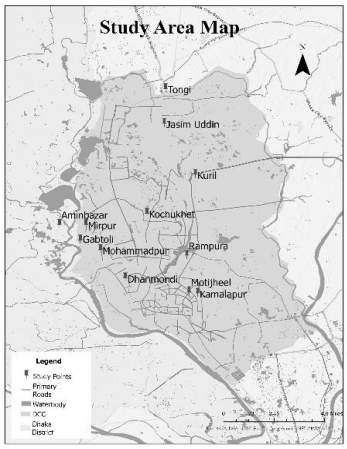
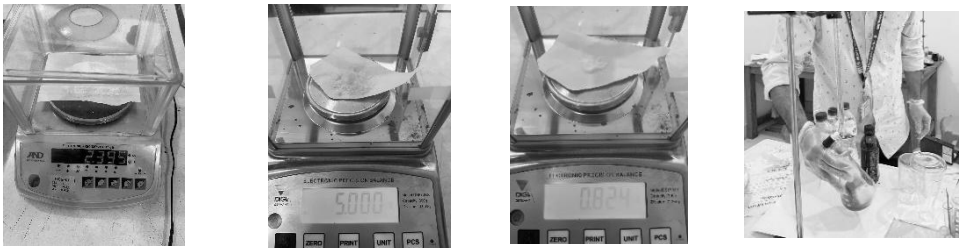


Figure 1: Location of water Sample collecting Point in This Study



Figure 2: Water sample



A B C D

Figure 3. Determination Chloride Ion in sample water: A. Weighing of $AgNO_3$(2.395 g), B. Weighing of K_2CrO_4, C. weighing of NaCl, D. Titration of the sample until the appearance of a permanent red-brown color



Figure 4. The color shifted from red-violet to purple, and finally to blue



A B

Figure 5: A. Start to disappear black spot at the bottom; B. Fully disappear at 90 mL

3. Result and Discussion

3.1 Sulfate

The measured sulfate concentrations in all the selected areas of Dhaka city ranged from <20 mg/L to a maximum of 90 mg/L in Amin Bazar as shown in Table 2, which is well below both the Bangladesh and WHO guideline value of 400 mg/L as shown in Table 3. According to standard classification, sulfate levels below 250 mg/L are considered safe for drinking and general domestic use. Most locations such as Jasim Uddin, Dhanmondi, Mohammadpur, Rampura, and others showed sulfate concentrations below 20 mg/L, indicating very low sulfate presence and suggesting minimal anthropogenic influence or effective protection of groundwater sources. Moderately higher values were observed in Mirpur (23 mg/L), Motijheel (27 mg/L), Kamlapur (42 mg/L), and Gabtoli (27 mg/L), likely due to moderate urban runoff and sewage contamination. Amin Bazar exhibited the highest sulfate level (90 mg/L), probably linked to landfill leachate, poor sanitation, and open dumping. Though all observed values fall within safe limits, elevated sulfate concentrations even below the threshold can contribute to scale buildup in pipelines, reduce the efficiency of water heaters, and slightly alter taste, affecting

DWASA water supply infrastructure and user experience. Prolonged exposure to higher sulfate levels in drinking water may also cause mild laxative effects, especially in children and the elderly. Continuous monitoring and improved waste management are essential to ensure long-term groundwater quality and protect public health.

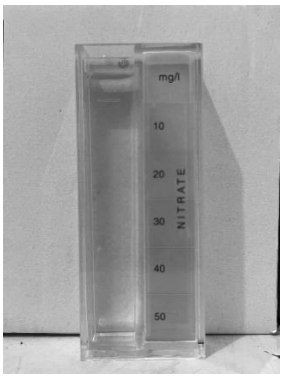


Figure: 6. The final color was visually compared with a standard reference scale.

Table 1: Classification Water Hardness according to WHO and Rand water

Total Hardness (mg/L)		Classification of Hardness	
WHO	Rand Water	WHO	Rand Water
<17	0-50	Soft	Soft
17-60	50-100	Slightly Hard	Moderately Soft
60-120	100-150	Moderately Hard	Slightly Hard
120-180	150-200	Hard	Moderately Hard
>180	200-300	Very Hard	Hard
-	>300	-	Very Hard

Table 2: Sulfate Ion, Chloride ion (Cl^-) and Nitrate Ion (NO_3^-) Concentrations in Groundwater Samples Collected from Various Locations in Dhaka City

Location	Sulfate ion (SO_4^{-2}) (mg/L)	Chloride ion (Cl^-) (mg/L)	Nitrate Ion (NO_3^-) mg/L
Tongi	<20	74	15
Jasim Uddin	<20	46	0
Abdullahpur	<20	64	0
Kuril	<20	42	0
Kochu Khet	<20	110	0
Rampura	<20	40	0
Mirpur	23	100	0
Motijheel	27	44	0
Kamlapur	42	52	0
Amin Bazar	90	234	20
Gabtoli	27	216	0
Dhanmondi	<20	54	0
Mohammadpur	<20	120	0

Table 3: Standard parameters for drinking water as per WHO and Bangladesh standards.

Parameter	Bangladesh Standards Value	WHO Guideline Value	Reference
Hardness (as $CaCO_3$)	150-500 mg/L	200 mg/L(aesthetic)	DPHE, 2015; WHO, 2017
Sulphate (SO_4^{2-})	400 mg/L	400 mg/L	DPHE, 2015; WHO, 2017
Chloride (Cl^-)	150-600 mg/L	≤ 250mg/L	DPHE, 2015; WHO, 2017
Nitrate (as NO_3^-)	≤10 mg/L (as N)	≤ 50mg/L(as NO_3^-)	DPHE, 2015; WHO, 2017

3.2 Hardness

The measured hardness of water samples from Dhaka city ranged from 33 mg/L at Kuril, classified *as* slightly hard according to WHO guidelines, to 300 mg/L at Amin Bazar, which falls into the very hard category as depicted in Table 1. Specifically, Kuril (33 mg/L) was slightly hard, Jasim Uddin (45 mg/L) and Motijheel (51 mg/L) were also in the slightly hard range, while Abdullahpur (60 mg/L) and Kochu Khet (90 mg/L) were classified as moderately hard as shown in Fig. 7. Locations like Tongi (126 mg/L), Mirpur (144 mg/L), Rampura (150 mg/L), Dhanmondi (105 mg/L), and Mohammadpur (150 mg/L) were in the hard category, while Gabtoli (195 mg/L) and Amin Bazar (300 mg/L) were in the very hard category. According to Bangladesh standards (150–500 mg/L) and WHO's aesthetic limit (200 mg/L), most samples were within acceptable limits, but Amin Bazar (300 mg/L) exceeded the WHO guideline, indicating potential problems such as scaling in plumbing

systems and reduced soap efficiency. These findings suggest significant spatial variability in water hardness across the city, reflecting differences in aquifer mineral content, and highlight the need for localized monitoring and appropriate treatment where necessary.

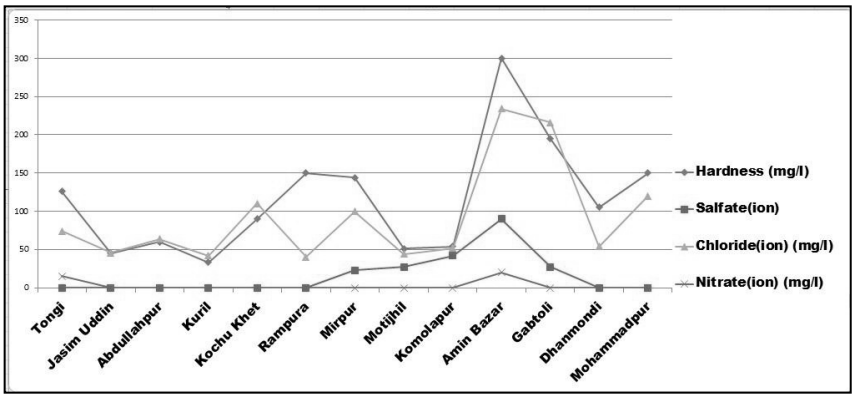


Figure 7: Concentrations of Hardness, Sulfate, Chloride, and Nitrate in DWASA supplied water

3.3 Chloride Ion (Cl^-)

The chloride ion (Cl^-) concentrations measured across various locations ranged from 39.99 mg/L to 233.93 mg/L as shown in Table 2. According to the World Health Organization (WHO), the guideline value for chloride in drinking water is 250 mg/L, primarily for aesthetic reasons such as taste and corrosivity. Bangladesh standards (BSTI) also recommend a range of 150–600 mg/L as depicted in Table 3. While all tested sites were within the permissible limits, Amin Bazar (233.93 mg/L) and Gabtoli (215.93 mg/L) were close to the upper boundary, indicating potential future risks. High chloride levels can corrode metal components in pumps, pipelines, and plumbing systems, leading to reduced equipment lifespan, leaks, and increased maintenance costs. Corrosion may also cause toxic metals like lead or iron to leach into the water supply, posing health risks. Although chloride itself is not harmful at these levels, the associated effects on infrastructure and secondary contamination are concerning. Areas like Mohammadpur and Kochukhet showed moderately high values and require regular monitoring. In contrast, locations such as Rampura, Kuril, and Motijheel exhibited low chloride levels and pose minimal risk. Preventive measures like using corrosion-resistant materials, regular testing, and public awareness are recommended to ensure long-term water safety and infrastructure integrity.

3.4 Nitrate (NO_3^-) concentrations

Nitrate (NO_3^-) concentrations in the sampled locations spanned 0 mg/L to 20 mg/L, with eleven sites including Jasim Uddin, Abdullahpur, Kuril, Gabtoli, Mohammadpur, Mirpur, Kamalapur, Motijheel, Dhanmondi, Rampura and Kochukhet showing no detectable nitrate, while Tongi registered about 15 mg/L and Amin Bazar about 20 mg/L as depicted in Table 2. These values are comfortably below the World Health Organization (WHO) guideline of 10 mg/L and the Bangladesh drinking-water standard of 50 mg/L as shown in Table 3, yet the two elevated sites warrant attention because nitrate can convert to nitrite in the body and impair oxygen transport, triggering methemoglobinemia in infants and posing long-term risks such as thyroid dysfunction. Rising nitrate often signals agricultural runoff, leaking sewage or aging plumbing that allows contaminated surface water to infiltrate supply lines; if unchecked, it can overload treatment units, foul pumps and hasten biofilm growth in distribution pipes. Consequently, Tongi and Amin Bazar should undergo more frequent monitoring, source-tracking and possibly improvements to wastewater control and pipe integrity, while the other locations remain low-risk provided routine surveillance and good agricultural and plumbing practices are maintained.

4. Conclusion

The groundwater quality assessment conducted across various locations of Dhaka city indicates that while the measured parameters including sulfate, hardness, chloride, and nitrate are within both the Bangladesh National Standards and WHO guideline values, spatial variations suggest localized sources of contamination and differing hydrogeological conditions. Amin Bazar consistently exhibited the highest concentrations for multiple parameters, particularly sulfate (90 mg/L), hardness (300 mg/L), chloride (233.93 mg/L), and nitrate (20 mg/L), pointing toward the influence of landfill leachate, poor sanitation infrastructure, and possible surface water intrusion. In contrast, residential areas such as Dhanmondi, Mohammadpur, and Jasim Uddin recorded lower contaminant levels, indicating relatively protected aquifers and effective groundwater abstraction practices. While none of the locations exceeded the permissible limits, elevated hardness in several areas (e.g., Gabtoli and

Amin Bazar) suggests potential issues with scale buildup in pipelines, reduced soap efficiency, and long-term infrastructure wear. Similarly, higher chloride values in Gabtoli and Amin Bazar may lead to corrosion of plumbing systems, indirectly posing risks through metal leaching. Though nitrate concentrations remained within safe limits, the slightly higher values in Amin Bazar and Tongi warrant attention, as they may indicate sewage leakage or agricultural runoff. These findings emphasize the need for area-specific groundwater management, regular quality monitoring, and improved urban waste disposal systems to safeguard the city's drinking water supply. Continuous surveillance and appropriate mitigation strategies are essential to maintain water quality, minimize public health risks, and ensure the longevity of DWASA's distribution infrastructure.

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